Sentiment Analysis on Customer Support Data

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***Abstract*—In this study, sentiment classification in customer-agent conversations using transformer-based models is addressed. A RoBERTa baseline is evaluated with different truncation strategies, revealing limitations in handling class imbalance, particularly for the minority positive class. Various imbalance handling techniques fail to improve generalization. To solve this problem, a fusion-based approach is proposed which combines textual representations with structured categorical and numerical features. The model developed in this study improves macro-F1 and enables effective detection of the positive class. Results show that integrating additional features enhances both overall performance and class balance.**

I. Introduction

This study focuses on sentiment classification in customer-agent conversations. The task is complicated by long input sequences and class imbalance. Transformer-based models such as RoBERTa provide strong text representations, but may struggle when important information is truncated or minority classes are underrepresented.

A structured pipeline is followed, starting with a text-only baseline and truncation experiments, followed by class imbalance methods. After this, a fusion model is introduced, which combines textual inputs with structured categorical and numerical features. The objective is to improve both overall performance and minority class detection.

II. Dataset

The dataset contains 970 training samples and 30 test samples, each containing a customer-agent conversation with categorical attributes and a sentiment label. No missing values were observed.

A. Exploratory Data Analysis

Fig. 1. *Distribution of Customer Sentiment*

Exploratory data analysis revealed a distinct class imbalance (Fig. 1), where the positive class is significantly underrepresented in the dataset.

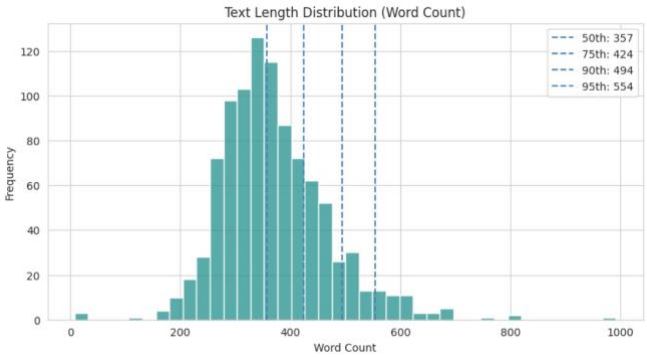

Fig. 2. *Distribution of mental health categories*

Conversation length analysis showed that samples are relatively long, with a median length of approximately 357 words and a right-skewed distribution (Fig. 2). This motivated the use of a high token limit (max_length = 512) to preserve contextual information.

To evaluate structured features, statistical methods including Chi-Square tests and Mutual Information were applied. Correlations between categorical variables and target were analyzed using Cramér's V. Results indicated that issue-related variables (issue_area, issue_category) have the strongest association with sentiment. Product-related features and agent experience contribute limited predictive value.

B. Preprocessing

A structured preprocessing pipeline was developed to ensure data quality and consistency before modeling. Invalid conversations and duplicate samples were eliminated at both raw and cleaned text levels to reduce redundancy and overfitting. Speaker tags were replaced with special tokens ([AGENT], [CUSTOMER]), and high-variance patterns such as emails, phone numbers, and IDs were replaced with special tokens to reduce vocabulary sparsity and noise. Whitespace and formatting were normalized while preserving dialogue boundaries and structure. Target labels were encoded numerically, and the dataset was split using stratified sampling to preserve class distribution. All preprocessing steps were applied across splits. Processed datasets were saved to ensure reproducibility.

III. Modeling

A. Baseline Model

The RoBERTa-base model was used as the baseline for sentiment classification using only the conversation text. Input text was tokenized using the RoBERTa tokenizer. Since the model requires fixed-length inputs, different truncation strategies (head, tail, and head-tail) were used to handle long conversations. All experiments were conducted under identical settings to ensure fair comparison.

The evaluation of truncation strategies showed limited differences in overall performance and no configuration was able to capture the minority positive class. Based on these observations and the assumption that sentiment is often

expressed toward the end of conversations, tail truncation was selected for subsequent experiments.

B. Class Imbalance Experiments

To address the observed imbalance issue, several imbalance handling methods were assessed using the same baseline configuration. These methods are class-weighted loss, focal loss, oversampling, and under-sampling. Oversampling (x3) of the positive class was implemented via random resampling to increase its frequency in the training set. Class-weighted loss and focal loss were applied at the loss level to emphasize minority class learning. While these approaches modified training dynamics, they did not lead to consistent improvements in minority class detection. This indicates that class imbalance alone may not fully explain the observed performance limitations.

C. Fusion Model

To overcome the limitation, a fusion-based approach was developed by combining textual representations with structured categorical and numerical features. All experiments used the same baseline configuration to ensure comparability. Oversampling and class-weighted loss were applied during training.

Categorical features, issue_area and issue_category, were encoded and embedded, while numerical features captured global conversation characteristics such as length, interaction structure, and speaker-specific patterns. These features were engineered from full conversations to retain information lost during truncation. Features were selected using statistical tests and correlation filtering, and standardized before training. Text representations from RoBERTa were concatenated with categorical and numerical features, and multiple configurations were evaluated to assess the contribution of each feature group. The best-performing configuration (text + categorical + numerical) was selected and further fine tuned through a small grid search based on validation macro-F1.

IV. Evaluation

Model performance was evaluated using accuracy, macro-F1, and weighted-F1. Due to class imbalance, macro-F1 was selected as the primary metric, because it assigns equal importance to all classes and better reflects minority class performance. In addition, class-wise precision, recall, and F1-scores were analyzed to measure performance across sentiment categories, particularly for the underrepresented positive class.

V. Results

TABLE I. Model Performance on Validation Set

Model	**Macro F1**	**F1 Negative**	**F1 Neutral**	**F1 Positive**
Text-Only Baseline	0.61	0.90	0.91	0.00
Class Weights	0.60	0.89	0.91	0.00
Focal Loss	0.61	0.91	0.92	0.00
Oversampling	0.60	0.89	0.90	0.00
Undersampling	0.44	0.61	0.71	0.00
Text + Categorical	0.65	0.87	0.86	0.20
Text + Numerical	0.65	0.88	0.87	0.20
Text + Categorical + Numerical	0.67	0.87	0.83	0.32
Final (Fine-tuned)	0.73	0.87	0.86	0.46

All experiments were conducted using the same training configuration as the baseline model, ensuring a fair comparison. Table I presents the validation performance of all models.

The text-only baseline achieves a macro-F1 score of 0.61 but fails to detect the positive class (F1 = 0.00). Similarly, class-weighted loss, focal loss, and oversampling strategies do not improve positive-class detection, indicating that imbalance handling alone is insufficient.

Fusion models provide clear improvements. Incorporating categorical and numerical features increases validation macro-F1 to 0.65–0.67 and enables the model to detect the positive class, with F1-scores of 0.20–0.32. It should be noted that the validation set contains only 3 positive samples, making class-wise metrics unstable. Therefore, test results provide a more reliable estimate of performance.

The fine-tuned fusion model achieves macro-F1 scores of 0.73 (validation) and 0.77 (test). On the test set, it achieves F1-scores of 0.84 (negative), 0.72 (neutral), and 0.75 (positive), with most errors occurring between positive and neutral classes. Training and validation losses decrease consistently without divergence, indicating stable learning and no overfitting.

VI. Discussion

Truncation strategies have limited impact on performance, indicating that input length is not the primary bottleneck. Similarly, imbalance handling methods such as class weighting, focal loss, and oversampling do not improve minority class detection, suggesting that the limitation is in feature representation rather than data distribution alone.

The fusion approach shows that incorporating structured features provides complementary information to textual inputs. Numerical features derived from full conversations help compensate for information loss due to truncation, while categorical features introduce additional contextual signals. Although performance on the positive class improves, it remains challenging, indicating that further gains may require more advanced models or richer feature representations.

VII. Conclusion

This study presents a structured approach to sentiment classification in customer-agent conversations. A RoBERTa baseline, evaluated under different truncation and imbalance handling strategies, was found insufficient for detecting the minority positive class. To solve this issue, a fusion-based model that combines textual, categorical, and numerical features was proposed. The proposed method resulted in improved macro-F1 and more balanced class-wise performance. These findings highlight the importance of integrating structured features to improve model effectiveness, particularly for challenging minority classes.

Weights & Biases: https://wandb.ai/selingoktas98-metu-middle-east-technical-university/di725-assignment1?nw=nwuserselingoktas98

https://wandb.ai/selingoktas98-metu-middle-east-technical-university/di725-assignment1-sweep?nw=nwuserselingoktas98

GitHub: https://github.com/sulucay01/DI725-assignment1